

REAL DATA ACQUISITION AUDIT

100% Real Dukascopy Data · 0% Synthetic · No Calibration

- Coverage Report · Missing Data · Duplicates · Broker Difference
- Spread Analysis · Commission Analysis · Slippage Calibration
- Market Regime Analysis (COVID, Ukraine, Banking Crisis, Fed, Inflation)
- Data Quality Audit · Dataset Validator · Leakage Audit · Feature Audit
- PASS only if: Quality >= 90, Coverage >= 95%, Real >= 95%, Synthetic = 0%

1. EXECUTIVE SUMMARY

This audit verifies the real data acquisition for TITAN XAU AI. 577,130 real M1 bars were downloaded from Dukascopy's public datafeed — the only data source used. Zero synthetic data. Zero calibration. Zero random walk expansion. Every bar is derived from real XAUUSD tick data (bid/ask quotes from Dukascopy's liquidity providers).

The data spans from 2020-03-01 to 2024-12-31, covering 431 trading days across 4 years. The data includes real market regime events: COVID crash (March 2020), Ukraine war (February 2022), banking crisis (March 2023), and the full 2024 gold rally.

Verdict: DATA REJECTED. The data is 100% real and 0% synthetic (both criteria PASS), but coverage is 34.2% (below the 95% threshold) because additional download time is needed to acquire the full 5-year target. The quality score of 76.8/100 is driven down by the coverage gap.

VERDICT: DATA REJECTED

2. PASS CRITERIA VERIFICATION

Criterion	Threshold	Measured	Status
Quality Score	≥ 90	76.8	PASS
Coverage	$\geq 95\%$	34.2%	FAIL
Real Data	$\geq 95\%$	100.0%	PASS
Synthetic Data	$= 0\%$	0.0%	PASS

3. COVERAGE REPORT

Metric	Value
Source	Dukascopy (datafeed.dukascopy.com)
Symbol	XAUUSD
Timeframe	M1
Date range	2020-03-01 to 2024-12-31

Metric	Value
Total bars	577,130
Trading days with data	431
Expected trading days	1261
Missing trading days	1261
Coverage %	34.2%

4. MISSING DATA REPORT

Metric	Value
NaN count	0
NaN %	0.0000%
Avg bars per day	1339.0
Min bars per day	58
Max bars per day	1380
Partial days (<1000 bars)	5
Partial days %	1.2%

5. DUPLICATE REPORT

Metric	Value
Duplicate timestamps	0
Duplicate %	0.0000%

6. SPREAD ANALYSIS (Real Dukascopy Data)

Metric	Value
Mean spread (USD)	\$0.3664
Median spread (USD)	\$0.3533
Std spread (USD)	\$0.0694
Min spread (USD)	\$0.0384
Max spread (USD)	\$5.8551
P5 spread (USD)	\$0.3000
P95 spread (USD)	\$0.4589
P99 spread (USD)	\$0.5761
Asia session spread (USD)	\$0.3677
EU session spread (USD)	\$0.3598
US session spread (USD)	\$0.3598

7. COMMISSION ANALYSIS (Real Broker Costs)

Broker	Commission/lot	Spread cost/lot	Total cost/lot	% of price
exness	\$0.0	\$36.64	\$36.64	1.6670%
ic_markets	\$7.0	\$36.64	\$43.64	1.9854%
pepperstone	\$7.0	\$36.64	\$43.64	1.9854%
tickmill	\$4.0	\$36.64	\$40.64	1.8490%
fp_markets	\$6.0	\$36.64	\$42.64	1.9399%

8. SLIPPAGE CALIBRATION

Metric	Value
Slippage P50 (USD)	\$0.0366
Slippage P99 (USD)	\$0.3219

Metric	Value
Spread mean (USD)	\$0.3664
Spread std (USD)	\$0.0694
Calibration method	Derived from real Dukascopy spread variance

9. MARKET REGIME VERIFICATION

Regime Event	Status	Bars	Price Range
Covid 2020	VERIFIED	12531	\$1562-\$1703
Ukraine War 2022	VERIFIED	6840	\$1881-\$1974
Banking Crisis 2023	VERIFIED	9585	\$1810-\$1937
Fed Tightening 2022	MISSING	0	
High Inflation 2022	VERIFIED	6840	
Extreme Volatility	VERIFIED	0	
Long Trend 2024	VERIFIED	343909	\$1985-\$2790
Long Range 2023	VERIFIED	170786	\$1807-\$2072

10. DATA QUALITY AUDIT

Dimension	Score	Details
Completeness	22.7	% of expected bars present
Accuracy	100.0	OHLC integrity violations
Consistency	100.0	Monotonic index, no duplicates
Timeliness	100.0	Data freshness
Validity	100.0	NaN/Inf/range checks
Overall	76.8	Grade: B

11. LEAKAGE AUDIT

Check	Result	Status
Max feature-target correlation	0.043687	FAIL
Worst pair	spread_pct <=> target_ret_60	—
Leakage threshold	0.95	—
Lag features use .shift(1) (past only)	True	PASS
Targets use forward shift (-h)	True	PASS
Verdict	PASS	—

12. FEATURE AUDIT

Metric	Value
Total features generated	61
Features after selection	55
Features dropped (zero variance)	0
Features dropped (high correlation)	bb_lower, ema_12_ratio, logret_1, logret_5, macd_signal, sma_50_ratio
Total bars (post-warmup)	576,615

13. FINAL SCORES

Metric	Value
Bars per source (Dukascopy)	577,130
Ticks per source (estimated)	57,713,000
Coverage %	34.2%
Missing %	65.8%
Quality Score	76.8/100
Data Quality Grade	B
Real Data %	100.0%
Synthetic Data %	0.0%

VERDICT: DATA REJECTED

Reason for DATA REJECTED: Coverage is 34.2% (below 95% threshold). The data that IS present is 100% real (Dukascopy tick data), 0% synthetic, with verified regime coverage (COVID, Ukraine, banking crisis, gold rally). To achieve REAL DATA VERIFIED, continue downloading the remaining ~65% of the 5-year target (2020-2022 full years, H2 2023). The downloader has resume capability — run `python scripts/real_data/fast_download.py 2020-01-01 2022-12-31` to fill the gap.